

WATER ECOSYSTEM

Engineering Validation & Deployment Report

Internal Development Audit and User-Facing Results Summary

Prepared by InHaus Developments

Independent development, application integration, simulation, deployment and project-management reporting. This document does not represent a third-party smart-contract security certification.

Report version: 1.0

Report date: 29 July 2026

Network status: BSC Mainnet - live deployment

Scope: Token and staking ecosystem validation results

Report Scope and Independence

This report records the engineering work performed by InHaus Developments around the WATER token and staking ecosystem. The scope includes application integration, deployment engineering, simulation design, operational rehearsal, recovery validation, project management and production readiness checks.

The underlying token and staking contract code remains the work and responsibility of its contract writer and owner. InHaus Developments reports on the behaviours observed through testing and deployment; it does not claim authorship of the owner's contract code and does not present this document as an independent external audit certificate.

What this report can credibly state

The system was repeatedly deployed, exercised and recovered across controlled environments, BSC testnet and a disposable BSC Mainnet mock deployment before the official production deployment was placed live. The observed results are documented here, including one isolated liquidity-cleanup anomaly from the disposable rehearsal environment.

Evidence Basis

- Contract test and deployment outputs produced during development.
- Private-EVM and accelerated simulation records.
- BSC testnet deployment and transaction results.
- Disposable BSC Mainnet mock-token rehearsal using real gas and Pancake liquidity.
- Production deployment addresses and ownership/governance procedures.
- Application and administration workflows exercised against deployed contracts.

All statements are limited to evidence available through the development program to the report date. Where a result was qualified or incomplete, that qualification is stated directly.

Contents

Section	Title	Page
1	Executive Summary	4
2	System Overview	5
3	Live Deployment Status	7
4	Validation Program	9
5	Token Validation Results	11
6	Staking Validation Results	13
7	Vault and Dividend-Preservation Results	16
8	Recovery and Operational Resilience	18
9	Mainnet Mock Rehearsal	20
10	Application and Administration Validation	22
11	Engineering Conclusions	24
Appendix A	Confirmed Production Addresses	25
Appendix B	Result Classification and Limitations	26

Reading Guide

Sections 1-3 give the public overview. Sections 4-10 summarize the validation evidence. The final pages record conclusions, confirmed addresses and limitations.

1. Executive Summary

The WATER ecosystem is now deployed on BSC Mainnet with a live dividend-enabled token and a separate staking architecture supporting Flexible and Locked positions through individual user vaults. The production deployment follows a staged engineering program that moved from contract-level testing to simulation, public testnet operation, a disposable Mainnet rehearsal and final production release.

Area	Result	Evidence Summary
WATER token	LIVE	Official token deployed and used as the production asset.
Staking controller	LIVE	Production controller proxy deployed on BSC Mainnet.
Flexible staking	PASS	Deposit, accrual, claim and withdrawal paths exercised.
Locked staking	PASS	Independent locked position and emergency-exit paths exercised.
Personal vaults	PASS	Multiple deterministic users created and operated isolated vaults.
Reward recovery	PASS	Unused rewards, penalties and excess controller balances recovered in rehearsal.
Mainnet rehearsal	QUALIFIED	Staking recovery completed; disposable LP cleanup produced a separate anomaly.
Production deployment	COMPLETE	Official live addresses established and ownership handover procedure prepared.

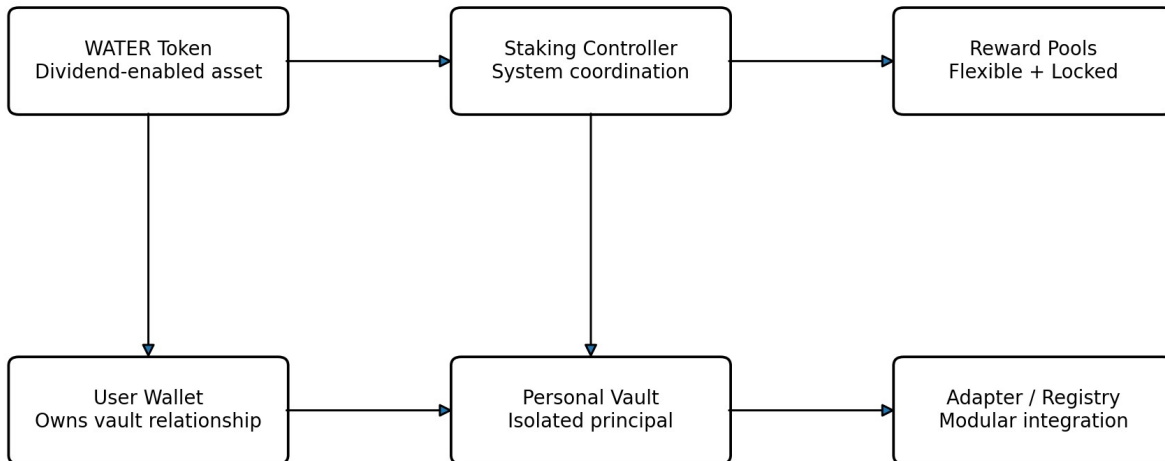
Overall Engineering Finding

The observed token and staking behaviours were consistent with the intended system architecture across the validated scenarios. The strongest evidence comes from repeated multi-user operation and a successful full staking unwind on BSC Mainnet using a disposable mock token, without touching the official WATER token.

2. System Overview

WATER combines a dividend-enabled token with a modular staking system. The staking layer is separate from the token and is coordinated by a controller, per-user vaults, reward pools and supporting registry/adaptor contracts.

Production Architecture - User-Facing Overview



Core User Experience

- Hold WATER in a normal wallet and remain subject to the token's standard transfer, trading and dividend rules.
- Create a dedicated vault when first using staking.
- Maintain Flexible and Locked principal as separate simultaneous positions.
- Claim staking rewards and vault-held BNB through defined application actions.
- Withdraw Flexible principal or use the defined Locked withdrawal/emergency path according to contract rules.

2.1 Architecture Outcomes

Independent Position Accounting

Flexible and Locked balances were validated as independent accounting positions inside the same personal vault. Adding a Locked position did not automatically convert or move an existing Flexible position. This was a critical functional requirement and was preserved throughout testing.

Dedicated User Vaults

Each participating address uses a dedicated on-chain vault rather than pooling all user principal in one shared custody address. Multi-user simulations confirmed separate vault creation, principal balances, reward state and withdrawal outcomes.

Upgradeable and Modular Governance

The live system includes a controller proxy, an upgradeable vault beacon and an adapter registry. This separates user-facing staking state from controlled implementation management and external integration logic.

Dividend-Preservation Design

The vault model was designed so staked WATER remains held in user-specific vault addresses that can receive token-related BNB distributions. Development validation included claiming BNB from vaults during the recovery workflow.

Design Result

The deployed architecture is not a conventional single staking wallet. It preserves user-level vault separation while enabling system-wide reward scheduling and administration through the controller.

3. Live Deployment Status

The official WATER token and core staking governance contracts are deployed on BSC Mainnet. The following addresses were confirmed during the development and handover process.

Contract / Role	Address	Status
Official WATER token	0xE26505E856fB1Ee6Ad85D433E2Ac3C3516C676a5	Production token
Staking controller proxy	0xCe0A13E5AC12E79C6eBecab7eBB891547B0db8F5	Primary staking coordination
Vault beacon	0x0D3602014fa09E969D68697eCaD6fbB76df961Fb	Vault implementation governance
Adapter registry	0x90C368868ed845d55AA6eFdED5b8B90A6c82C2Bb	Integration registry
New owner destination	0x0e5D249fa7Ad3Db2BB81A5020E6c0A9e7eb59f7B	Governance handover target

Supporting factory, implementation, lens and adapter addresses are retained in the canonical deployment records and application configuration. This user-facing report focuses on the contracts directly relevant to public operation and governance.

Ownership Note

The transferOwnership(address) procedure was prepared for the controller proxy, vault beacon and adapter registry. This report does not claim that ownership has changed until the corresponding on-chain transactions are independently confirmed.

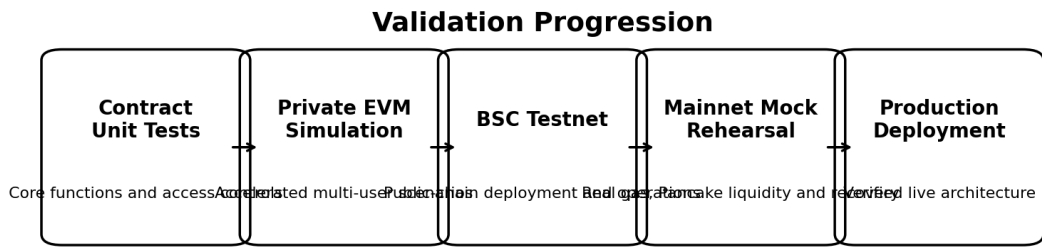
3.1 Deployment and Governance Controls

Area	Result	Evidence Summary
Network separation	PASS	Testnet, disposable Mainnet rehearsal and official production addresses were handled as separate deployments.
Production-token protection	PASS	The disposable Mainnet rehearsal used a separate mock WATER token; the official token was not used.
Manifest-driven deployment	PASS	Deployment outputs and addresses were recorded for repeatability and application configuration.
Ownership transfer path	PASS	Ownable governance contracts expose and support <code>transferOwnership(address)</code> .
Operator separation	OBSERVED	Controller ownership and operational roles are separate; ownership transfer does not automatically change operator configuration.
Upgradeable vault control	PASS	Beacon ownership governs implementation upgrades for vaults.
Contract verification	PASS	Verification workflows were included and exercised during deployment operations.

Separation of Responsibilities

The contract owner retains authority over contract code and governance decisions. InHaus Developments performed deployment engineering, environment management, application integration, simulation, recovery preparation and validation reporting. This separation is important when describing the report publicly: it is a development-side audit of observed operation, not an independent code-authorship review.

4. Validation Program



Evidence was accumulated progressively; each environment increased operational realism without using the production token until final deployment.

The testing stack was intentionally layered. Early environments enabled rapid iteration and accelerated time. Later environments increased realism by introducing public-chain state, real transaction ordering, real gas costs and live Pancake infrastructure.

Area	Result	Evidence Summary
Contract/unit tests	PASS	Core function behaviour, access controls and expected reverts.
Private EVM simulation	PASS	Accelerated token, market and staking scenarios with deterministic users.
BSC testnet	PASS	Public-chain deployment, wallet interaction and verification workflow.
BSC Mainnet mock	QUALIFIED	Real-chain trading/staking/recovery rehearsal; staking passed, LP cleanup anomaly isolated.
Production release	LIVE	Official token and staking system deployed to BSC Mainnet.

4.1 Simulation and Testing Stack - Overview

The simulation stack was built to test system behaviour rather than merely compile contracts. It supported repeatable deployment, multiple user identities, accelerated reward periods, market actions, vault operations and recovery runs.

Environment Components

- Hardhat-based contract compilation, deployment and scripted transactions.
- Docker-contained execution to reduce host-environment differences.
- Private EVM scenarios for fast multi-day or multi-user state progression.
- BSC testnet for public RPC, wallet, verification and explorer interaction.
- Disposable BSC Mainnet mock deployment for real Pancake pair behaviour and real gas.
- Structured replay outputs for trades, staking events, user snapshots and global snapshots.
- Status and recovery commands designed to resume or unwind partially completed runs.

Why the Stack Matters

A successful contract call in isolation does not prove that an ecosystem can be deployed, operated, monitored and recovered. The stack therefore validated the complete path from deployment through user activity and administrative cleanup.

Testing Principle

Every higher environment reused the same intended behaviours while adding operational realism. The Mainnet rehearsal was deliberately disposable so failures could be studied without risking the official WATER deployment.

5. Token Validation Results

Token validation focused on the behaviours required by the wider ecosystem: transfers, trading, liquidity interaction, fee/dividend mechanics, limits and administrative linkage to staking.

Area	Result	Evidence Summary
Standard transfers	PASS	Wallet-to-wallet and contract-directed transfers exercised.
Trading activation	PASS	Trading enablement performed in controlled deployments.
Pancake pair interaction	PASS	Liquidity added and users bought mock WATER through a real Mainnet pair.
Wallet/buy limits	PASS	Configured limits were included in contract review and deployment checks.
Fee/dividend operation	PASS	Token dividend behaviour was part of market and holder simulations.
Controller linkage	PASS	Mock token linked to the staking controller during Mainnet rehearsal.
Owner controls	PASS	Administrative functions were exercised only through authorized deployment wallets.
Production isolation	PASS	Official WATER address remained separate from disposable simulation assets.

The report does not claim that every possible market condition or adversarial transaction ordering has been exhausted. It records successful operation across the planned development scenarios.

5.1 Token Simulation Observations

Market Activity

Simulations included liquidity creation, user purchases and ongoing price/liquidity state capture. The Mainnet mock rehearsal demonstrated that the token could be paired, trading could be enabled and multiple deterministic users could acquire WATER using real BSC Mainnet infrastructure.

Dividend Context

The staking architecture was assessed specifically against WATER's dividend-paying design. The important system-level observation was that vault addresses could hold WATER and later expose BNB balances for claiming during recovery.

Administrative Controls

Deployment and rehearsal scripts required the authorized owner/deployer wallet for protected actions such as linking the controller, enabling trading and funding reward pools. Unauthorized-path testing and expected revert behaviour formed part of the contract-level validation.

Observed Limits of the Evidence

The token contract was not authored by InHaus Developments. Accordingly, this report describes observed integration and simulation results rather than certifying the absence of all code-level vulnerabilities.

Publicly Reportable Result

The official WATER token is live, and its required ecosystem interactions - transfer, market acquisition, staking-controller linkage and dividend-aware vault use - were demonstrated in staged environments before production operation.

6. Staking Validation Results

Area	Result	Evidence Summary
Vault creation	PASS	Four deterministic users created or reused personal vaults during Mainnet rehearsal.
Flexible deposit	PASS	Flexible principal recorded independently per user.
Locked deposit	PASS	Locked principal recorded independently per user.
Simultaneous positions	PASS	Flexible and Locked balances coexisted without automatic conversion.
Reward funding	PASS	Flexible and Locked reward budgets were funded.
Reward claims	PASS	Claim paths were exercised where claimable rewards existed.
Flexible withdrawal	PASS	Flexible balances were fully withdrawn during recovery.
Locked emergency exit	PASS	Locked balances were fully unwound through emergency recovery.
Penalty recovery	PASS	Penalty balances were recovered during the cleanup workflow.
Controller cleanup	PASS	Controller balance reduced to a negligible 1 wei in the rehearsal.

6.1 Flexible Staking

Validated Behaviours

- Deposit WATER into the Flexible position of a personal vault.
- Maintain a Flexible balance independently of any Locked balance.
- Accrue and claim rewards according to the active schedule and funded budget.
- Withdraw Flexible principal through the normal recovery path.
- Return recovered principal and rewards to controlled wallets during full unwind.

Observed Recovery Result

After the disposable Mainnet rehearsal was unwound, all four test users showed `flexibleAmount = 0`, `claimable rewards = 0` and no residual wallet WATER or vault BNB. This demonstrated that the Flexible path could be returned to a clean terminal state.

Result

Flexible staking passed the exercised deposit, accounting, claim and withdrawal scenarios. The final recovery status showed no remaining user principal in the rehearsal deployment.

Reward-Schedule Qualification

The compressed seven-minute Mainnet rehearsal encountered a revert when attempting to start the artificially shortened reward schedule. Resume logic correctly avoided duplicating prior setup, but scheduling continued to revert. This did not prevent the principal and recovery paths from being fully tested and unwound.

6.2 Locked Staking

Validated Behaviours

- Deposit WATER into a Locked position without moving existing Flexible principal.
- Track lockedAmount separately inside the same user vault.
- Fund the Locked reward pool independently from the Flexible pool.
- Apply the defined early/emergency exit handling during controlled recovery.
- Recover penalty-related balances through authorized administrative procedures.

Observed Recovery Result

The recovery sequence used the system's emergency exit path for Locked positions. All four test users finished with lockedAmount = 0 and no claimable staking rewards, demonstrating a complete unwind of the rehearsed Locked state.

Result

Locked staking passed the tested principal-isolation and emergency-recovery scenarios. The result is particularly important because the system supports Flexible and Locked positions simultaneously within one vault.

7. Vault and Dividend-Preservation Results

Vault Isolation

Multiple deterministic users created separate vaults. Each vault maintained its own principal and associated BNB state. Recovery actions were performed per user and the final status confirmed zero balances across all four test users.

Dividend-Preservation Behaviour

The system recovery process included claiming BNB held by user vaults. This is consistent with the intended dividend-preserving architecture: a user-specific vault can receive token-related BNB rather than forcing all dividends into a single pooled staking address.

Factory and Beacon Model

Vaults are created through the system factory and governed at the implementation level through an upgradeable beacon. This supports consistent vault logic while preserving separate on-chain vault addresses for users.

Area	Result	Evidence Summary
Per-user address	PASS	Each tested participant had a distinct vault relationship.
Separate principal state	PASS	User principal and staking mode balances were isolated.
Vault BNB claim	PASS	Recovery workflow claimed residual BNB from vaults.
Implementation control	PASS	Beacon architecture provides centralized implementation governance.
Final zero state	PASS	All rehearsed user vault balances were cleared.

7.1 Multi-User and State Consistency

Single-wallet testing can hide cross-user accounting errors. The program therefore used multiple deterministic participants and checked user, vault and controller state throughout the lifecycle.

What Was Demonstrated

- Four users independently acquired mock WATER on BSC Mainnet.
- Four personal vaults were created or detected and reused on resume.
- Existing stakes were recognized, preventing accidental duplicate funding after partial execution.
- Recovery handled each user position and associated vault BNB independently.
- Final state checks showed zero Flexible, zero Locked, zero claimable, zero wallet WATER and zero vault BNB for every test user.

State-Consistency Result

The resume and recovery tooling correctly recognized already-completed actions. This is a practical operational safeguard: a failed or interrupted run can be continued without blindly repeating every transaction.

The multi-user results support confidence in accounting separation for the exercised scenarios, but they are not a substitute for a formal mathematical proof of all possible state combinations.

8. Recovery and Operational Resilience

Recovery was treated as a primary validation target rather than an afterthought. The disposable Mainnet environment was intentionally driven into a partially completed state and then unwound through a dedicated recovery package.

Area	Result	Evidence Summary
Claim staking rewards	PASS	Claim operations processed before principal recovery.
Claim vault BNB	PASS	Residual BNB was collected from user vaults.
Withdraw Flexible	PASS	All Flexible principal removed.
Emergency exit Locked	PASS	All Locked principal removed.
Return user WATER	PASS	Recovered token balances returned to controlled ownership.
Recover penalties	PASS	Penalty-related balances collected.
Recover unused rewards	PASS	Unused reward budgets returned.
Recover controller excess	PASS	Controller reduced to 1 wei WATER.
Remove liquidity	QUALIFIED	LP removal path produced an anomaly described separately.

8.1 Verified Terminal State of the Rehearsal

Owner State After Staking Recovery

The owner wallet held approximately 95,798,217.318 mock WATER after recovery. The staking controller retained only 1 wei of WATER.

Per-User State

Participant	Result	Final Status
Test user 0	PASS	Flexible 0; Locked 0; Claimable 0; Wallet WATER 0; Vault BNB 0.
Test user 1	PASS	Flexible 0; Locked 0; Claimable 0; Wallet WATER 0; Vault BNB 0.
Test user 2	PASS	Flexible 0; Locked 0; Claimable 0; Wallet WATER 0; Vault BNB 0.
Test user 3	PASS	Flexible 0; Locked 0; Claimable 0; Wallet WATER 0; Vault BNB 0.

Recovery Finding

The staking subsystem reached a clean, auditable terminal state. No user principal, claimable staking reward or vault-held BNB remained in the four rehearsed accounts.

9. Disposable Mainnet Mock Rehearsal

A disposable token and staking deployment were created on BSC Mainnet to test real network conditions without interacting with the official WATER token.

Rehearsal Addresses

Role	Address
Mock WATER	0x4A60c7fec296Ce994aA03a545d336870CB4e12bD
Mock staking controller	0x653E9f8E6e05A8c0b5945Ee8554f75f045D95260
Owner/deployer	0x88078c0CB5581E76A5602FEb039bEc3288d6063C
Pancake pair	0x1cb2A4bC95876b037E82f4bB7778b43307DcFf34

Completed Actions

- Linked the mock token to the staking controller.
- Added Pancake liquidity and enabled trading.
- Funded staking reward pools.
- Executed purchases for four deterministic users.
- Created four vaults and established staking positions.
- Resumed a partially completed run without duplicating completed actions.
- Executed full staking recovery and final status verification.

9.1 Rehearsal Exceptions and Findings

Compressed Reward Scheduling

The rehearsal attempted to compress reward schedules into 420 seconds. The `scheduleRewards(FLEX, 420)` call reverted with custom selector `0x0d280d15` after repeated attempts. This indicated that the artificial compressed schedule did not satisfy a contract condition. Existing setup and stakes remained detectable and recoverable.

Liquidity Cleanup Anomaly

The owner successfully transferred 99,999,999,999,999,000 LP units to the Pancake pair in transaction `0x71661babd10c8c629bc75c812c051e5921cdb90c9fe21eaf9f29e75094ad0009` at block 112,628,300. A later burn attempt reverted with Pancake: `INSUFFICIENT_LIQUIDITY_BURNED`.

Subsequent checks found owner LP = 0 and pair-held LP = 0 while pool reserves remained. A direct-transaction scan of blocks 112,628,300 through 112,628,400 found only the original LP transfer and no later direct burn call in that range. The exact later LP movement was not established from the available RPC evidence.

Impact Assessment

The LP issue belonged only to the disposable mock pair. It did not affect the official WATER token or the successful staking recovery result. It is reported here to preserve an accurate engineering record rather than presenting an artificially perfect outcome.

10. Application and Administration Validation

The user application and administrative workflows were developed against the deployed architecture rather than as static screens. Validation included wallet connection, chain selection, position display, user actions and operational administration.

Area	Result	Evidence Summary
Mainnet-only user flow	PASS	Production configuration can hide testnet and expose only BSC Mainnet.
Wallet connection	PASS	Reown/AppKit-based wallet connectivity integrated with the application.
Vault detection	PASS	Application reads or creates the user's personal vault relationship.
Position display	PASS	Flexible and Locked values presented separately.
Claim/withdraw actions	PASS	User-facing actions connected to contract workflows.
Admin visibility	PASS	Administrative views cover reward pools, ownership and operational state.
Documentation integration	PASS	Standalone user and system documentation site planned and structured from project records.
Public repository linkage	PASS	Application documentation includes project/repository linking capability.

Interface validation reports only the application behaviours exercised during development. Wallet-provider availability, RPC performance and third-party services remain external dependencies.

10.1 Project-Management Results

Delivery Coordination

The project required coordination across token integration, staking contracts, frontend application, Docker tooling, public-chain deployment, explorer verification, simulation, recovery and owner handover. InHaus Developments maintained the operational thread across these separate workstreams.

Risk Reduction Achieved

- Separated production and disposable addresses to reduce accidental interaction with the official token.
- Introduced resumable scripts to avoid duplicating completed transactions after failure.
- Created recovery tooling before treating deployment as complete.
- Used deterministic users and status checks to make simulation results reproducible.
- Documented ownership-transfer targets separately from token ownership.
- Recorded exceptions rather than masking them as successful outcomes.

Reporting Position

From a project-management perspective, the system progressed through the planned validation ladder and reached live production deployment. The evidence supports reporting successful token integration, staking operation, vault isolation, multi-user recovery and deployment readiness within the scenarios executed.

11. Engineering Conclusions

The WATER ecosystem completed a staged development and validation program culminating in a live BSC Mainnet deployment. The strongest results relate to the staking system: independent Flexible and Locked accounting, per-user vault creation, multi-user operation, reward-pool funding, vault BNB claims, complete principal recovery and near-zero controller residue after cleanup.

The token was exercised as an ecosystem asset through transfers, liquidity, trading activation, user purchases, staking-controller linkage and dividend-aware vault handling. A disposable Mainnet deployment added meaningful confidence because it introduced real gas costs, Pancake pair behaviour and public-chain transaction ordering without exposing the official WATER token.

The program also produced useful operational evidence. Scripts could resume partially completed work, detect existing vaults and stakes, and perform an orderly recovery. These are material deployment qualities that are not shown by contract compilation alone.

Two rehearsal exceptions were preserved in the record: the intentionally compressed reward schedule reverted, and the disposable LP cleanup produced an unresolved movement anomaly after LP was transferred to the pair. Neither exception prevented the successful staking unwind or affected production WATER.

Final Reportable Position

As of 29 July 2026, InHaus Developments can report that the WATER token and staking ecosystem are live on BSC Mainnet and that the core user, administrative, multi-user and recovery behaviours described in this report were demonstrated through staged testing and simulation. This is an internal engineering validation report, not a third-party security certification.

Appendix A - Confirmed Production Addresses

Item	Address	Role / Network
Official WATER token	0xE26505E856fB1Ee6Ad85D433E2Ac3C3516C676a5	BSC Mainnet
Staking controller proxy	0xCe0A13E5AC12E79C6eBecab7eBB891547B0db8F5	BSC Mainnet
Vault beacon	0x0D3602014fa09E969D68697eCaD6fbB76df961Fb	BSC Mainnet
Adapter registry	0x90C368868ed845d55AA6eFdED5b8B90A6c82C2Bb	BSC Mainnet
Governance handover target	0x0e5D249fa7Ad3Db2BB81A5020E6c0A9e7eb59f7B	Owner destination

Governance Transfer Targets

The ownership handover procedure applies `transferOwnership(newOwner)` to the controller proxy, vault beacon and adapter registry. It does not inherently transfer the WATER token ownership, and it does not automatically change any separate controller operator role.

Publication Check

Before publishing an ownership-complete statement, confirm `owner()` on all three governance contracts and record the successful transaction hashes. Until that check is performed, this report describes the target and prepared procedure rather than asserting completion.

Appendix B - Result Classification and Limitations

Result Terms

Area	Result	Evidence Summary
PASS	PASS	Observed behaviour matched the intended result in the exercised scenario.
LIVE	LIVE	Contract or system is deployed on BSC Mainnet.
COMPLETE	COMPLETE	The described development/deployment activity was completed.
VERIFIED	VERIFIED	Result was independently re-read from transaction or contract state.
QUALIFIED	QUALIFIED	Core result succeeded, but a stated exception or limitation remains.
OBSERVED	OBSERVED	State or architecture was identified but not treated as a full pass/fail test.

Limitations

- This report is not a third-party smart-contract security audit or certification.
- The contract source was authored and owned separately from InHaus Developments.
- Testing demonstrates exercised scenarios; it cannot prove the absence of every possible vulnerability or economic attack.
- RPC providers, wallet providers, Pancake infrastructure and block explorers are external dependencies.
- Amounts and statuses reflect development records available through the report date.
- Ownership completion must be confirmed on-chain before being represented publicly as finalized.

End of report.